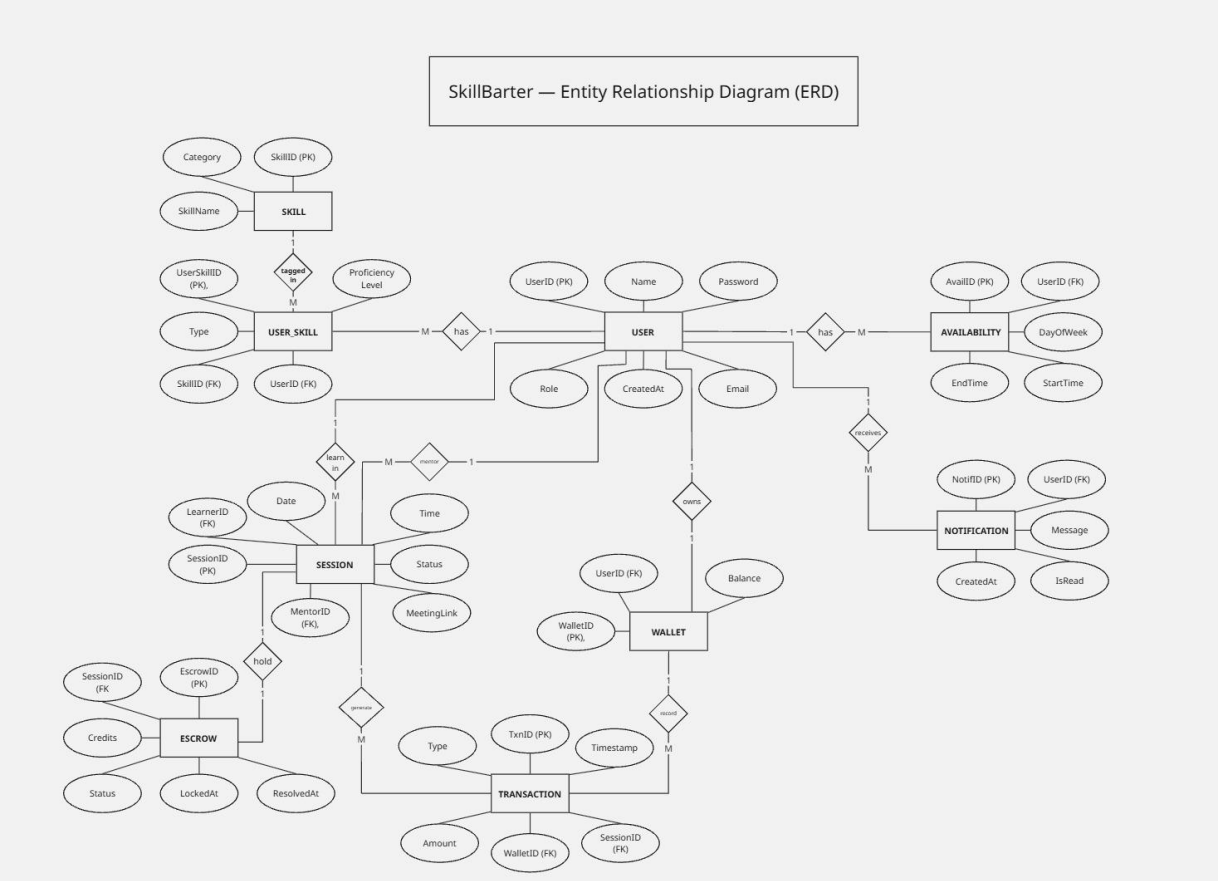
2.1 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram below shows all the data entities in SkillBarter, their attributes, and how they are related to each other. The system uses MongoDB Atlas as the database, with 9 main collections. The ERD follows standard notation — single line for 1, crow's foot for Many.



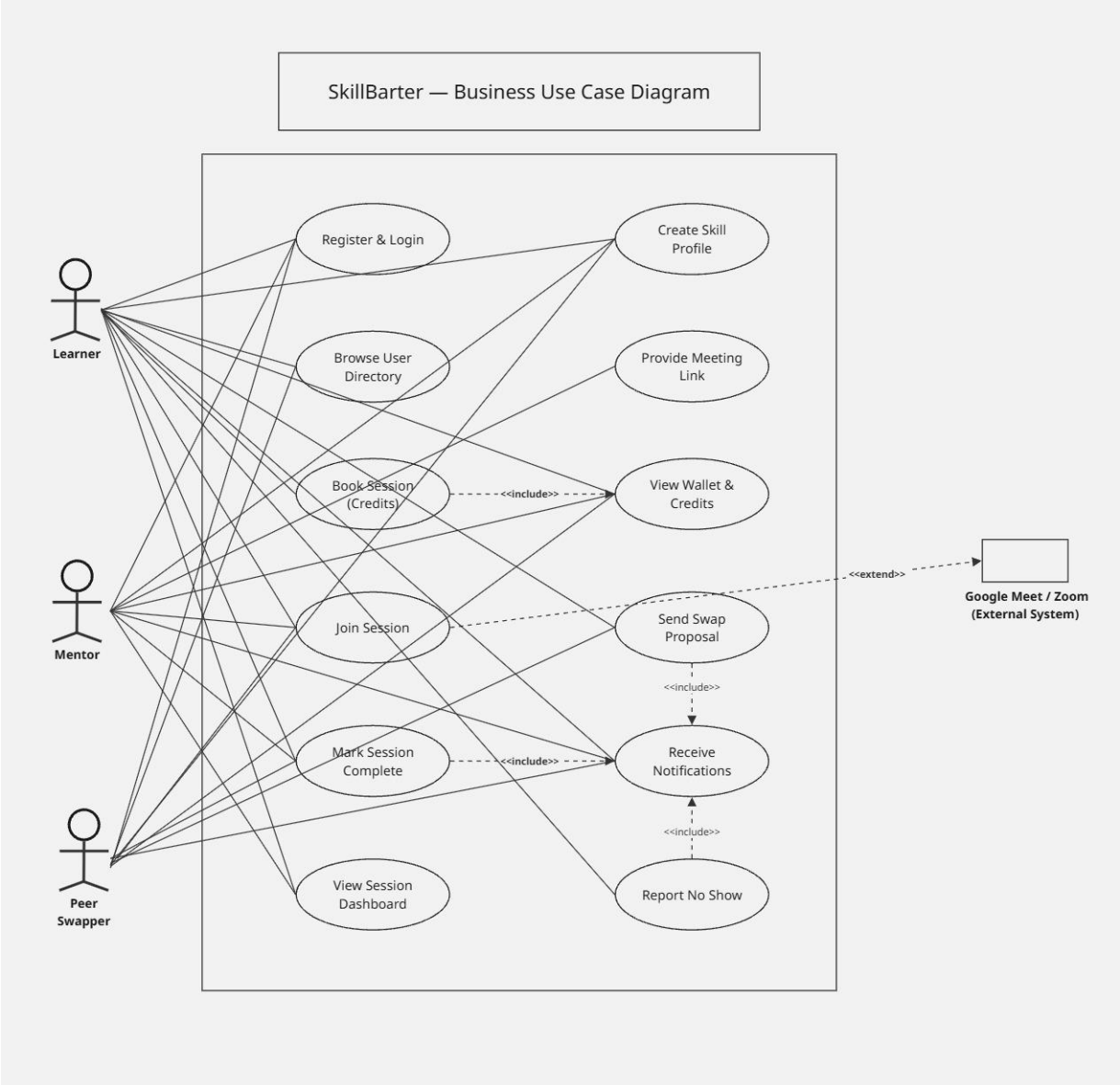
Entities and Relationships Summary:

Entities	Cardinality	Description
USER — USER_SKILL	**1 : N**	One user can have many skill entries (offered or wanted)
SKILL — USER_SKILL	**1 : N**	One skill can appear in many user skill entries
USER — AVAILABILITY	**1 : N**	One user can have many availability time slots
USER — SESSION (Mentor)	**1 : N**	One mentor can conduct many sessions
USER — SESSION (Learner)	**1 : N**	One learner can attend many sessions
SESSION — ESCROW	**1 : 1**	Each session has exactly one escrow record
USER — WALLET	**1 : 1**	Each user has exactly one wallet
WALLET — TRANSACTION	**1 : N**	One wallet has many transaction entries

SESSION — TRANSACTION	**1 : N**	One session can generate multiple transactions
USER — NOTIFICATION	**1 : N**	One user receives many notifications

2.2 Business Use Case Diagram

The Business Use Case Diagram below shows the interactions between the three main actors (Learner, Mentor, Peer Swapper) and the external system (Google Meet / Zoom) with the SkillBarter platform's use cases.



Actors:

Actor	Role Description
Learner	A registered user who books sessions using Skill Credits to learn from mentors.
Mentor	A registered user who teaches skills and earns Skill Credits per session.
Peer Swapper	A registered user who both teaches and learns via direct skill swap (no credits needed).
Google Meet / Zoom	External video conferencing platform used to conduct the live session.

2.3 Table Structure / Data Dictionary

The following section describes the complete data dictionary for all 9 collections in the SkillBarter MongoDB Atlas database. Each table shows field names, data types, constraints, and descriptions.

Table 1: USER Collection

Field Name	Data Type	Constraint	Description
UserID	ObjectId	PK, Auto	Unique identifier for each user
Name	String	Required	Full name of the user
Email	String	Required, Unique	User's email address for login
Password	String	Required	bcrypt hashed password
Role	Enum	Required	Mentor / Learner / PeerSwapper
CreatedAt	DateTime	Auto	Account creation timestamp

Table 2: SKILL Collection

Field Name	Data Type	Constraint	Description
SkillID	ObjectId	PK, Auto	Unique identifier for the skill
SkillName	String	Required, Unique	Name of the skill (e.g. React, Python)
Category	String	Required	Category (e.g. Web Dev, Design, Data)

Table 3: USER_SKILL Collection

Field Name	Data Type	Constraint	Description
UserSkillID	ObjectId	PK, Auto	Unique identifier for this entry
UserID	ObjectId	FK → USER	References the user
SkillID	ObjectId	FK → SKILL	References the skill
Type	Enum	Required	offer (teach) or want (learn)
ProficiencyLevel	Enum	Required	Beginner / Intermediate / Expert

Table 4: AVAILABILITY Collection

Field Name	Data Type	Constraint	Description
AvailID	ObjectId	PK, Auto	Unique identifier
UserID	ObjectId	FK → USER	References the user
DayOfWeek	String	Required	Monday, Tuesday... Sunday
StartTime	String	Required	Available from (e.g. 10:00 AM)
EndTime	String	Required	Available until (e.g. 12:00 PM)

Table 5: SESSION Collection

Field Name	Data Type	Constraint	Description
SessionID	ObjectId	PK, Auto	Unique identifier for the session
MentorID	ObjectId	FK → USER	References the mentor
LearnerID	ObjectId	FK → USER	References the learner
SessionDate	Date	Required	Scheduled date of the session
SessionTime	String	Required	Scheduled time of the session
MeetingLink	String	Optional	Google Meet or Zoom link (added by mentor)
Status	Enum	Required	Pending / Confirmed / Completed / Cancelled
SessionType	Enum	Required	Swap (no credits) or Credit (1 credit)

Table 6: ESCROW Collection

Field Name	Data Type	Constraint	Description
EscrowID	ObjectId	PK, Auto	Unique identifier for the escrow record
SessionID	ObjectId	FK → SESSION	References the booked session
Credits	Number	Required	Amount of credits locked (usually 1)
Status	Enum	Required	Locked / Released / Refunded / Forfeited
LockedAt	DateTime	Auto	Timestamp when credit was locked
ResolvedAt	DateTime	Optional	Timestamp when escrow was resolved

Table 7: WALLET Collection

Field Name	Data Type	Constraint	Description
WalletID	ObjectId	PK, Auto	Unique identifier for the wallet
UserID	ObjectId	FK → USER, Unique	References the user (one wallet per user)
Balance	Number	Default: 0	Current available Skill Credit balance

Table 8: TRANSACTION Collection

Field Name	Data Type	Constraint	Description
TxnID	ObjectId	PK, Auto	Unique identifier for this transaction
WalletID	ObjectId	FK → WALLET	References the wallet
SessionID	ObjectId	FK → SESSION	References the related session
Type	Enum	Required	Earn / Spend / Refund / Forfeit
Amount	Number	Required	Number of credits in this transaction
Timestamp	DateTime	Auto	Date and time of the transaction

Table 9: NOTIFICATION Collection

Field Name	Data Type	Constraint	Description
NotifID	ObjectId	PK, Auto	Unique identifier for the notification
UserID	ObjectId	FK → USER	User who receives the notification
Message	String	Required	Notification message text
IsRead	Boolean	Default: false	Whether the user has read it
CreatedAt	DateTime	Auto	Timestamp when notification was created